JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A Place to Learn, A Chance to Grow"

Education to Eternity

Session: 2025-26

Progress Seminar No. - 02

Title: Implementation of multi - model coding tutor and collaboration platform by using Generative AI

Under the Guidance of Prof. Anuja Ghasad
Designation: Assistant Professor, JDCOEM, Nagpur

Presented by :
1) Arya Thool (3CSE-19)
2) Kajal Khandekar (3CSE-46)
3) Arya surkar (3CSE-20)
4) Abhishek Gupta (3CSE-04)

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Introduction

- **Programming** forms the foundation of computer science education, but many students **face persistent difficulties** in mastering the underlying **logic of problem-solving**.
- Traditional resources such as textbooks, tutorials, and coding platforms often present concepts in **isolated formats**, making learning **fragmented** and **less effective**.
- Students frequently struggle to translate **ideas** or **diagrams** into **executable code**, which **slows down the learning process** and **reduces confidence.**
- This results in a significant gap between **theoretical understanding** and **practical application** of programming concepts.

Introduction

- **LogicFlow AI** is an advanced code synthesis platform that teaches programming logic using text, voice, diagrams, and collaborative tools.
- A semantic logic understanding engine that explains reasoning step-by-step and translates user intent into code.
- Built-in real-time collaborative editor with pair-programming, live hints, and in-session assessments.

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
1	Designing an intelligent tutoring system for computer programing in the Pacific	Priynka Sharma, Mayuri Harkishan	BMC Medical Education 2022	Proposes a programming tutor architecture using optimized algorithms, aiming to accelerate programming skills and provide personalized learning via intelligent, interactive modules.
2	AI Powered Code Reviewer	Shaik Nagur Vali, M. Vikas, V. Smaran, E. Augusteen, N. Varun	International Journal of Research Publication and Reviews 2024	MERN stack application where submitted code is reviewed using AI, offering suggestions on syntax errors, code quality enhancement, and developer productivity.
3	Software engineering education in the era of conversational AI	C Sengul et al.	PLOS Digital Health 2024	Analyzes conversational AI agents' support in teaching requirements, software design, and quality assurance, highlighting educational benefits and practical implementation challenges.

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
4	AI-powered Code Review with LLMs: Early Results	Z Rasheed et al.	arXiv 2024	Describes a multi-agent LLM system for code review, improving bug detection, optimization feedback, and bridging gaps in traditional code analysis.
5	Collaborative Coding Platform	Sangamesh Dhannur, et al	International Journal of Creative Research Thoughts 2025	A real-time, multi-user platform for live code synchronization, collaborative programming activities, and effective communication, enhancing remote teamwork and learning outcomes.
6	A systematic review of AI-powered collaborative learning in higher education	A Kovari et al	Computers & Education: Artificial Intelligence 2025	Systematically examines AI-enhanced collaborative learning, detailing integration strategies, effectiveness, and impact on peer-to-peer knowledge sharing in higher education.

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
7	Large Language Models in Computer Science Education: A Systematic Literature Review	Nishat Raihan, Mohammed Latif Siddiq, Joanna C.S. Santos, Marcos Zampieri	VL/HCC 2023	Reviews use of LLMs in computer science education, evaluating improved personalization, student engagement, and instructional efficacy, while noting remaining deployment challenges.
8	Multimodal AI in Enhancing Intelligent Tutoring Systems	ESPIN Group Research	ESPIN Group 2025	Explores how integrating text, speech, and visual cues in ITS enables adaptive, responsive instruction and deeper student engagement with diverse learning modalities
9	Exploring the Role of AI Assistants in Computer Science Education	T Wang et al	Computers & Education: Artificial Intelligence 2025	Evaluates ChatGPT's performance on computer science problems and gathers instructor perspectives, informing future AI assistant deployment for effective student support.

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
10	Generative AI and Its Impact on Personalized Intelligent Tutoring Systems	Subhankar Maity, Aniket Deroy,	arXiv 2024	Examines LLM integration in tutoring for dynamic learning paths, immediate feedback, and improved adaptive educational experiences.
11	Towards Educator-Driven Tutor Authoring: Generative AI Approaches for Creating Intelligent Tutor Interfaces	Tommaso Calò, Christopher J. MacLellan,	arXiv 2024	Shows how generative AI simplifies tutor interface creation; empowers teachers to craft adaptive, user-driven tutoring tools.
12	AI-Powered Pedagogical Frameworks: Advancing Computational Thinking and Coding Proficiency in Secondary Education	Lin, Kuan Yeh; Po-Hsun Cheng; Li-Wei Chen	Research Square Preprint 2025	Describes AI-driven inquiry learning; Socratic questioning fosters computational thinking and coding proficiency through adaptive interactions.

Education to Eternity

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
13	A Collaborative Code Platform with Advanced AI Features and Real-Time Collaboration Tools	Aarti Gaikwad, Anudip Gupta, et al.	International Journal for Research in Applied Science & Engineering Technology 2024	Combines real-time collaboration, advanced AI text analysis, and multilingual support for dynamic, globally-accessible coding teamwork.
14	Survey of Natural Language Processing for Education: Taxonomy, Systematic Review, and Future Trends	Yunshi Lan, Xinyuan Li, et al.	arXiv 2024	Reviews NLP breakthroughs for educational use: automated assessment, feedback, and language models for personalized instruction and analytics.
15	Conversational AI in Education: A General Review of Chatbot Technologies and Challenges	Wasin Alkishri, Jabar H Yousif, Yousuf Nasser Al Husaini, Mahmood Al-Bahri	International Journal for Research in Applied Science & Engineering Technology 2025	PRISMA-driven review of educational chatbots, highlighting their effectiveness in formative assessment, tutoring, and administrative support.

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
16	Computer Science Education in the Age of Generative AI	Russell Beale	arXiv 2025	Explores generative AI's impact on teaching coding, presenting policy suggestions for effective, rigorous integration into CS education.
17	A Systematic Review of Generative AI for Teaching and Learning Practice	Bayode Ogunleye, et al.	arXiv 2024	Synthesizes 355 studies showing generative AI improves teaching and learning; recommends deeper classroom and curricular integration.
18	AICodeReview: Advancing Code Quality with AI-enhanced Review Processes	Y. Almeida et al.	Computers in Human Behavior Reports 2024	Researches AI-based code review tools for automated assessment and feedback, streamlining code quality assurance.

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
19	A Comprehensive Review of AI-based Intelligent Tutoring Systems: Applications and Challenges	Meriem Zerkouk, Miloud Mihoubi, Belkacem Chikhaoui	arXiv 2024	Systematic review mapping intelligent tutoring advances—pedagogical, NLP, and adaptive learning techniques—with current successes and persistent issues.
20	A Systematic Review of AI-driven Intelligent Tutoring Systems in K-12 Education	A. Létourneau et al.	PLOS Digital Health 2024	Evidence-based overview of AI's impact on school learning, covering improvements in student outcomes and experimental design insights.

Research Gap

- **Multimodal Integration Deficiency:** Most current studies emphasize text or single-modality AI tutoring systems; there is limited exploration of fully integrated multimodal approaches combining text, speech, visuals, and interaction modalities to enhance coding education.

- **Real-Time, Context-Aware Feedback:** There is a research gap in providing real-time, context-sensitive, AI-generated feedback during live coding and collaborative sessions that adapts dynamically to users' skill levels and interaction patterns.

- **Ethical and Bias Considerations:** Research on mitigating AI bias, ensuring fairness in AI-driven coding tutors, and addressing data privacy within collaborative AI learning environments remains underexplored.

Research Gap

- **Limited Evaluation of Generative AI Pedagogy in Coding:** There is sparse empirical research assessing the long-term pedagogical effectiveness and learning outcomes of generative AI-driven coding tutors, particularly regarding student engagement, retention, and skill transfer.

- **Scalability and Performance under Collaboration:** Research lacks focus on the scalability of AI tutoring systems under concurrent multi-user load in real-time collaborative coding platforms.

- **Explainability and Transparency of AI Decisions:** There is a lack of focus on making AI tutor reasoning and decisions interpretable to learners and instructors, which is critical for trust, learning comprehension, and user acceptance.

Statement of the problem

- Learning programming is not just about writing code, but about developing the ability to **think logically**.
- Most existing learning platforms focus either on **syntax practice** or **problem-solving in isolation**, leaving students struggling to connect concepts with real-world coding.
- The absence of **instant**, **contextual feedback** makes it harder to correct mistakes and build confidence.
- Current tools provide **limited support for real-time collaboration**, which is crucial for peer-to-peer learning and teamwork.
- This creates a significant gap between **understanding programming concepts** in theory and **applying them effectively** in practice, highlighting the need for an integrated solution.

Objectives of the study

- To design a **multi-modal** learning platform that integrates **text**, **voice**, and **diagram-based inputs** for teaching programming logic.
- To develop a **semantic logic understanding engine** that explains code step-by-step and bridges the gap between conceptual design and implementation.
- To provide **instant**, **contextual feedback** that helps learners debug, correct mistakes, and build programming confidence.
- To implement a **real-time collaborative coding environment** that supports peer-to-peer learning and mentorship.
- To validate the platform through **usability testing**, **feedback collection**, and **benchmarking** against existing tools.

Scope

- The project focuses on **building a multi-modal coding tutor** that supports text, voice, and diagram-based inputs for learning programming logic.

- It will include a **semantic engine** capable of translating user intent and diagrams into code, along with step-by-step logical explanations.

- The platform will feature a **real-time collaborative editor** to enable pair programming, live hints, and teamwork among learners.

- The study will cover **usability testing**, **performance benchmarking**, and **peer validation** against conventional coding platforms.

- The scope is limited to **teaching programming logic and collaboration** rather than full-fledged automated software development.

Limitations

- **Complexity of Semantic Understanding:** Accurately translating diagrams, natural language, or voice inputs into precise code logic is a challenging task.
- **Integration Complexity:** Combining multiple modes (text, voice, diagrams) into a single seamless workflow may lead to integration issues.
- **Resource Requirements:** Running AI models for semantic parsing and feedback may require high computational resources, limiting scalability on low-end systems.
- **Real-Time Collaboration Challenges:** Ensuring seamless synchronization, low latency, and conflict resolution in collaborative coding can be technically demanding.

Methodology

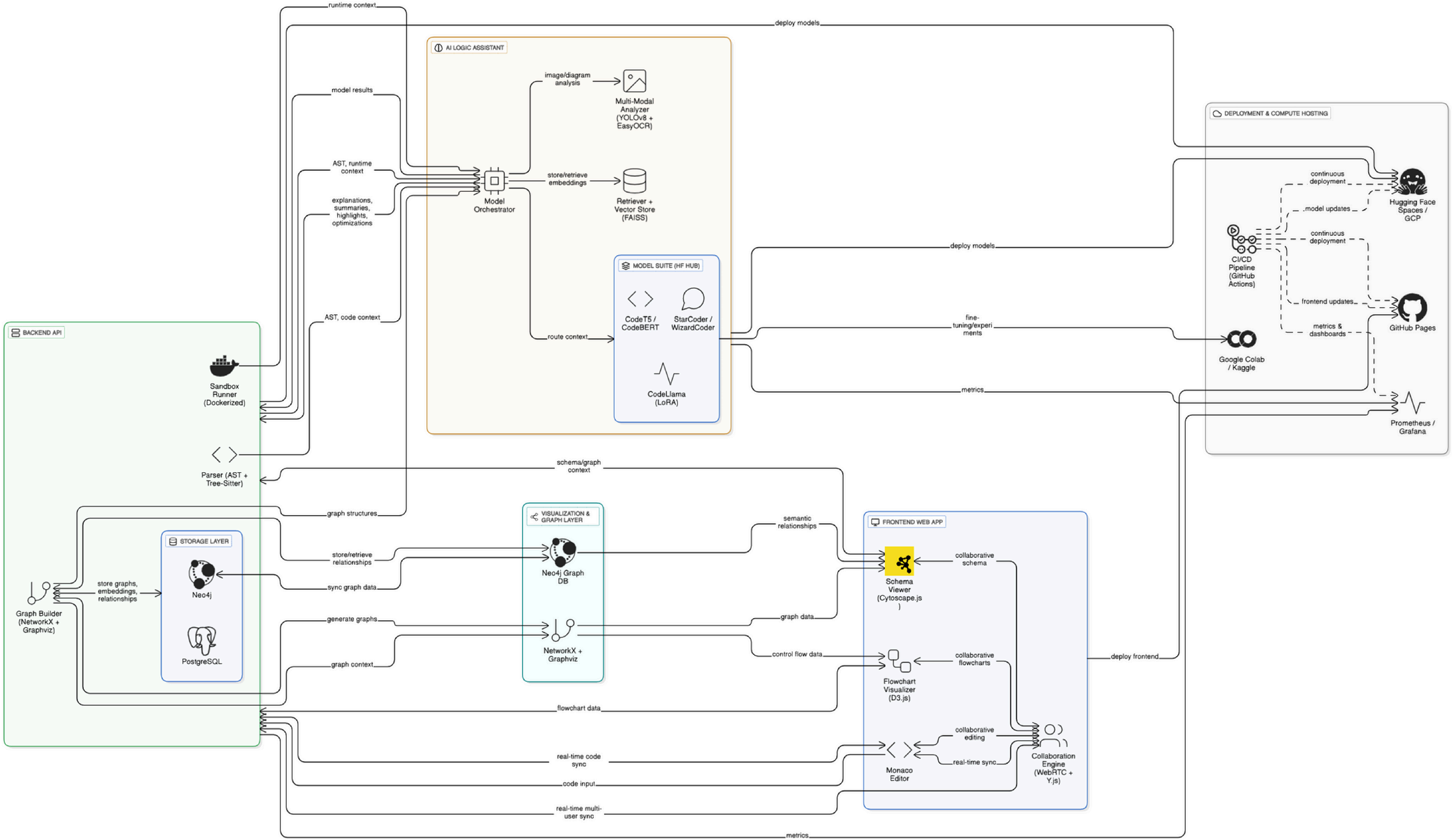
- **Exploratory Research:** Conduct a comprehensive literature review on **program synthesis**, **AI compilers** and **collaborative debugging** approaches.
- **Agile Prototyping:** Follow an incremental development cycle, beginning with **AST parsing** and **flowchart visualization**, and progressively integrating **AI assistance**, and **collaboration features**.
- **Human-Centric Design:** Focus on explainable AI that emphasizes **logic assistance** over auto-coding, supports **skill-building**, and leverages **visual schema** for effective learning.

Methodology

- **Evaluation & Validation:** Perform **usability studies**, **peer feedback sessions**, and **benchmarking** against existing tools (e.g., GitHub Copilot).

- **Deployment & Dissemination:** Deploy the final platform via GitHub/Hugging Face for accessibility, supported by a demo presentation and a potential IEEE paper submission.

Methodology (Flowchart)



Tools

- **Frontend:** React.js, D3.js, Cytoscape.js (flowchart/schema visualization), WebRTC + Y.js (collaboration).
- **Backend:** Python (FastAPI/Flask), AST (Python stdlib), tree-sitter (multi-language parsing).
- **Visualization & Graphs:** Graphviz, NetworkX, Neo4j (knowledge graphs).
- **AI/ML Models: 1. Code Understanding:** CodeT5, CodeBERT.
  2. Logic Explanation: StarCoder, WizardCoder.
  3. Optimization: CodeLlama, fine-tuned with LoRA.

Tools

- **Other AI Tools:** YOLOv8 + EasyOCR (multi-modal input from images/diagrams).
- **Compute & Hosting:** Google Colab, Kaggle, Google Cloud Free Credits, Hugging Face Spaces, GitHub Pages.
- **Collaboration & Docs:** GitHub (version control, project site), VS Live Share (inspiration for real-time debugging).

Techniques

- **Code Parsing & Representation:** Abstract Syntax Tree (AST), semantic embeddings, knowledge graph construction.
- **Visualization:** Flowchart and schema generation via graph-based rendering, drag-and-drop visual logic building.
- **Collaboration Techniques:** Real-time synchronization (Y.js), WebRTC-based multi-user debugging, shared logic templates.
- **Performance Optimization:** Quantization, ONNX runtime, caching frequent AI outputs.

Techniques

- **Evaluation Techniques:** Usability testing, peer surveys, complexity analysis (Big-O, bottleneck prediction).

AI Techniques:

- Transformers for semantic code explanation.
- Fine-tuning via LoRA for domain-specific optimization.
- Explainable AI methods (attention weight highlighting, context mapping).

Work Plan

Phase 1 (Months 1–2): Research & Foundation

- Literature survey, finalize problem statement, collect datasets (CodeSearchNet, GitHub repos).
- Build POC: Python AST parsing → basic tree output.

Phase 2 (Months 3–4): Core Prototyping

- Implement code-to-flowchart (AST → Graphviz → JSON → D3.js).
- File schema visualizer (import/dependency graphs).
- React UI: code editor + visualization panels.

Phase 3 (Months 5–7): AI Logic Assistance

- Integrate CodeT5, CodeBERT, StarCoder.
- Build “Explain my code” feature: line-by-line + function-level NL explanations.
- Tooltip integration with flowcharts + schema AI summaries.

Work Plan

Phase 4 (Months 8–9): Collaboration & Refinement

- Add WebRTC/Y.js for multi-user coding sessions.
- Collaborative debugging with shared flowchart/schema updates.
- Model optimization for faster inference.

Phase 5 (Months 10–11): Evaluation & Paper Writing

- Conduct surveys on comprehension improvement.
- Draft IEEE paper: problem, solution, architecture, experiments.
- Usability tests + bug fixes.

Phase 6 (Month 12): Final Demo & Submission

- Record demo video: Code → Flowchart → Schema → AI Explanation → Collaboration.
- Submit final report, IEEE paper, GitHub repo, and licensing.

Implications

- **Academic Impact:** Enhances students' ability to understand programming logic by providing **multi-modal explanations** and **hands-on practice**.

- **Skill Development:** Promotes **teamwork**, **problem-solving**, and **debugging skills** through interactive and collaborative learning.

- **Research Contribution:** Opens avenues for further exploration in **multi-modal AI tutors**, **semantic code analysis**, and **human–AI collaboration** in education.

Conclusions (expected)

- The study anticipates that **LogicFlow AI** will bridge the gap between **learning concepts** and **applying them in code**, addressing challenges in programming.
- The project is expected to demonstrate the feasibility of **integrating multimodal learning** (text, voice, diagrams) **with real-time collaboration** in a single platform.
- By focusing on **explainability** and **logic assistance**, the platform will encourage deeper understanding rather than surface-level coding.
- **LogicFlow AI** will serve as a foundation for building more **interactive**, **explainable**, and **student-friendly** coding tools in academic environments.

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Thank You!!!